Hyperspectral Geology - Postdoctoral Researcher

Lawrence Livermore National Laboratory · Livermore, CA, United States

Job ID (this archive)	ENERGY-115
Employer	Lawrence Livermore National Laboratory
Location(s)	Livermore, CA, United States
Career stage (archive tag)	Graduate degree / research (Postdoctoral)
Location group	Other U.S.
Majors (archive tags)	Materials Science & Engineering, Data Science, Computational & Geospatial Science
Energy area	Critical minerals / hyperspectral mapping
Pay listed	Not listed
Posted / listing date	46317.790717488424
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://jobs.smartrecruiters.com/LLNL/3743990016352206-hyperspectral-geology-postdoctoral-researcher
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-086

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

Company Description

Join us and make YOUR mark on the World!

Lawrence Livermore National Laboratory (LLNL) has turned bold ideas into world-changing impact advancing science and technology to strengthen U.S. security and promote global stability.

Our mission spans four critical national security areas nuclear deterrence, threat preparedness, energy security, and multi-domain defense empowering teams to take on the toughest challenges of today and tomorrow. With a culture built on innovation and operational excellence, LLNL is a place where your expertise can make a real impact.

Job Description

Lawrence Livermore National Laboratory's Postdoctoral Research Program provides mentored training to enhance professional skills and research independence for individuals who have recently earned their doctoral degree and are pursuing their chosen career path.

We have an opening for a Postdoctoral Researcher to investigate mineral assemblages and geochemical signatures in legacy mine tailings as indicators of critical-mineral potential. Candidates from mineralogy, geology, geochemistry, mining, hyperspectral imaging, multimodal data science, machine learning, and/or related disciplines are encouraged to apply.

This position is in the Atmospheric, Earth, and Energy Division, within the Physical and Life Sciences Directorate.

You will

- Combine whole-tailings chemistry, hyperspectral and other quantitative imaging, and complementary characterization data to identify abundant mineral proxies and assemblages associated with trace critical minerals.
- Evaluate relationships among site geology, ore-processing history, mineral alteration, weathering, and spectral response, with a major focus on identifying the visible and near-infrared to short-wavelength infrared (VNIR-SWIR) features and wavelength bands most useful for recognizing critical-mineral proxy assemblages under laboratory and field conditions.
- Examine how mineral mixing, grain size, moisture, surface roughness, and weathering influence spectral detection and uncertainty.

- Design and implement statistical, chemometric, and/or machine learning and Super Intelligence (formerly referred to as Artificial Intelligence, or AI) techniques and reproducible quantitative workflows in one or more of the above areas.
- Actively participate in conception, design, and execution of independent (but complementary) research and collaborate with a broad spectrum of scientists (geologists, mineralogists, microscopists, spectroscopists, data scientists, physicists, and engineers developing field-deployable characterization systems) internally and externally to the Laboratory to achieve project goals.
- Organize, analyze, and present results at external conferences, seminars and/or technical meetings; and publish research results in peer-reviewed scientific or technical journals.
- Perform other duties as assigned.

Qualifications

- PhD in geology, geochemistry, mining, or a related field.
- Experience with one or more of the following areas: quantitative mineralogical or geochemical characterization; hyperspectral imaging or reflectance spectroscopy; scanning electron microscopy/energy dispersive spectroscopy (SEM-EDS) imaging, or multimodal data integration.
- Experience with quantitative characterization of mine deposits, tailings, waste rock, or other heterogeneous geological materials.
- Experience with VNIR-SWIR hyperspectral imaging.
- Analytical proficiency with python/pytorch, MATLAB, R, machine learning packages, or comparable tools.
- Proficient verbal and written communication skills to collaborate effectively in a team environment; demonstrated ability to explain technical information; and record of independent research and publication.
- Effective initiative and interpersonal skills and ability to work in a collaborative, multidisciplinary team environment.

Qualifications we desire

- Additional experience with spectral calibration, spectral unmixing, feature extraction, spectral library development, reflectance spectroscopy, and/or reduced-band sensor design.
- Experience with machine learning (e.g., image segmentation, spatial analysis, chemometrics, clustering, classification, probabilistic modeling) or Super Intelligence (formerly referred to as Artificial Intelligence, or AI) methods for proxy identification, uncertainty quantification, or detection of mineral associations.
- Experience with multimodal fusion of chemical, mineralogical, spectral, image, and geospatial data.
- Experience with additional characterization techniques such as X-ray diffraction, X-ray fluorescence, Raman spectroscopy, UV-excited fluorescence, etc.

Pay Range

$123,048 Annually

This is the lowest to highest salary we in good faith believe we would pay for this role at the time of this posting. An employee's position within the salary range will be based on several factors including, but not limited to, specific competencies, relevant education, qualifications, certifications, experience, skills, seniority, geographic location, performance, and business or organizational needs.

Additional Information

Position Information

This is a Postdoctoral appointment with the possibility of extension to a maximum of three years. The postdoctoral appointment is a temporary, mentored training position intended for individuals in the early stages of their postdoctoral research careers. Consistent with the structure of the program, applicants must have received their Ph.D. (or equivalent doctorate) within five years of the anticipated start date.

Why Lawrence Livermore National Laboratory?

- Included in 2026 Best Places to Work by Glassdoor!
- Flexible Benefits Package
- 401(k)
- Relocation Assistance
- Education Reimbursement Program
- Flexible schedules (*depending on project needs)
- Our values - visit https://www.llnl.gov/inclusion/our-values

Security Clearance

None required. However, if your assignment is longer than 179 days cumulatively within a calendar year, you must go through the Personal Identity Verification process. This process includes completing an online background investigation form and receiving approval of the background check.

National Defense Authorization Act (NDAA)

The 2025 National Defense Authorization Act (NDAA), Section 3112, generally prohibits citizens of China, Russia, Iran and North Korea without dual US citizenship or legal permanent residence from accessing specific non-public areas of national security or nuclear weapons facilities. The restrictions of NDAA Section 3112 apply to this position. To be qualified for this position, Candidates must be eligible to access the Laboratory in compliance with Section 3112.

Pre-Employment Drug Test

External applicant(s) selected for this position must pass a post-offer, pre-employment drug test. This includes testing for use of marijuana as Federal Law applies to us as a Federal Contractor.

Wireless and Medical Devices

Per the Department of Energy (DOE), Lawrence Livermore National Laboratory must meet certain restrictions with the use and/or possession of mobile devices in Limited Areas. Depending on your job duties, you may be required to work in a Limited Area where you are not permitted to have a personal and/or laboratory mobile device in your possession. This includes, but not limited to cell phones, tablets, fitness devices, wireless headphones, and other Bluetooth/wireless enabled devices.

If you use a medical device, which pairs with a mobile device, you must still follow the rules concerning the mobile device in individual sections within Limited Areas. Sensitive Compartmented Information Facilities require separate approval. Hearing aids without wireless capabilities or wireless that has been disabled are allowed in Limited Areas, Secure Space and Transit/Buffer Space within buildings.

How to identify fake job advertisements

Please be aware of recruitment scams where people or entities are misusing the name of Lawrence Livermore National Laboratory (LLNL) to post fake job advertisements. LLNL never extends an offer without a personal interview and will never charge a fee for joining our company. All current job openings are displayed on the Career Page under “Find Your Job” of our website. If you have encountered a job posting or have been approached with a job offer that you suspect may be fraudulent, we strongly recommend you do not respond.

To learn more about recruitment scams:
https://www.llnl.gov/sites/www/files/2023-05/LLNL-Job-Fraud-Statement-Updated-4.26.23.pdf

Equal Employment Opportunity

We are an equal opportunity employer that is committed to providing all with a work environment free of discrimination and harassment. All qualified applicants will receive consideration for employment without regard to race, color, religion, marital status, national origin, ancestry, sex, sexual orientation, gender identity, disability, medical condition, pregnancy, protected veteran status, age, citizenship, or any other characteristic protected by applicable laws.

Reasonable Accommodation

Our goal is to create an accessible and inclusive experience for all candidates applying and interviewing at the Laboratory. If you need a reasonable accommodation during the application or the recruiting process, please use our online form to submit a request.

California Privacy Notice

The California Consumer Privacy Act (CCPA) grants privacy rights to all California residents. The law also entitles job applicants, employees, and non-employee workers to be notified of what personal information LLNL collects and for what purpose. The Employee Privacy Notice can be accessed here.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: Lawrence Livermore National Laboratory’s Postdoctoral Research Program provides mentored training to enhance professional skills and research independence for individuals who have recently earned their doctoral degree and are pursuing their chosen career path. | PhD in geology, geochemistry, mining, or a related field. | This is a Postdoctoral appointment with the possibility of extension to a maximum of three years. The postdoctoral appointment is a temporary, mentored training position intended for individuals in the early stages of their postdoctoral research careers. Consistent with the structure of the program, applicants must have received their Ph.D. (or equivalent doctorate) within five years of the anticipated start date.

Key skills listed in the archive: Python; MATLAB; R; PyTorch; machine learning; hyperspectral

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer SmartRecruiters Employer / ATS identifier: REF9009W

Career stage: Postdoctoral

Major fit: Advanced research pathway: geology, hyperspectral imaging and geospatial data; PhD required. Energy-sector connection: Hyperspectral mapping of critical-mineral potential in mine tailings.

Retrieval note: New posting found in expansion round. Official SmartRecruiters job data reports active=true and supplies a job-specific apply URL. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer SmartRecruiters. Source identifier: REF9009W.

Sources and retrieval details

Employer posting

https://jobs.smartrecruiters.com/LLNL/3743990016352206-hyperspectral-geology-postdoctoral-researcher

Retrieved source

https://api.smartrecruiters.com/v1/companies/LLNL/postings/3743990016352206

Retrieved: 2026-10-26T16:35:32.639748+00:00 (UTC). Major fit, career stage and energy-sector connection are archive annotations; the employer text above is preserved from the cited source.